



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Consult 4: CA6121 AI in Project Management

Group Project Progress Report Writing Progress and Tips

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iPhone 17 Pro Max
256GB



2ND



Samsung Galaxy S25+ 8GB RAM 256GB

3RD



11-inch iPad Wi-Fi 256GB

4TH & 5TH



Sony Wireless
Noise Cancellation
Headphones

6TH & 7TH



Garmin Forerunner
55 Smartwatch

8TH to 52ND



SGD \$30
Grab Voucher

Recap



Tutorial Activity

AI in Project Management



Tutorial Activity/ Exercise: 03

Recap

Topic:

Schedule & Critical Path: AI Tools implementation in Project Management.

This is **INDIVIDUAL Exercise**.

You are a project manager and planning to implement some improved solutions to your work processes / workflow for the project management .



Tutorial Activity/ Exercise: 03

Recap

Activity:

Work as an **INDIVIDUAL** on the exercises in the class.

Tools used:

ChatGPT, Make, Excel, Google Drive or similar of student's choice.

Problem statement:

You have received the client documentation and need to be sorted out to prepare a business case. (Document provided as per the instructions).

Please use the AI Tools which you can use for your project.



Tutorial Activity/ Exercise: 03

Recap

PART 1, 2: (90min)

Part 1: (60 min)

- i. Use the prompt provided to generate the result using AI Tool of your choice.
- ii. Perform some research and evaluate what are the best tools do you wish to use.



Tutorial Activity/ Exercise: 03

Recap

PART 1, 2: (90min)

Part 2: (60 min)

- i. Create a workflow mentioning the tools you proposed / used.
- ii. Mention maximum five (5) challenges you faced and how did you overcome these?
- iii. Do you think, this workflow is helpful for the project managers? Please write 5 positive and 5 negative points.
- iv. Upload the results in the folder provided in NTU Learn.

Quick Recap



Cost prediction and forecasting

Project Cost Management



Earned Value Management (EVM)

Quick Recap

Earned Value Management (EVM) is a **methodology that combines scope, schedule, and resource measurements to assess project performance and progress.**

It integrates scope baseline and cost baseline, along with the schedule baseline to form the performance management.



Earned Value Management (EVM)

Earned Value Management (EVM) develops and monitor 3 key dimensions for each work package and control account:

- 1. Planned Value (PV)**
- 2. Earned Value (EV)**
- 3. Actual Cost (AC)**



Earned Value Management (EVM) Example

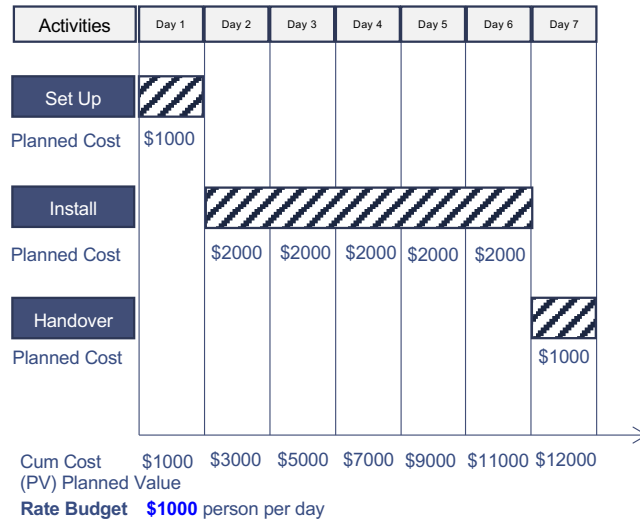
A company plan to install new computers supply and install by the supplier as follows:

Day	Work	Person required (no.)	Planned Rate (\$)	Planned Cost (\$)
1	Setup	1	\$1000	\$1000
2	Installation	2	\$1000	\$2000
3	Installation	2	\$1000	\$2000
4	Installation	2	\$1000	\$2000
5	Installation	2	\$1000	\$2000
6	Installation	2	\$1000	\$2000
7	Final handover	1	\$1000	\$1000

How do we set this information to monitor?



Earned Value Management (EVM) Example



This was the Plan
But what happened in actual?



Earned Value Management (EVM) Example

ACTUAL Scenario: Company **could not find person at planned rates:**

Work done **1st day is 100%** and **2nd day 50% by 2 people.**

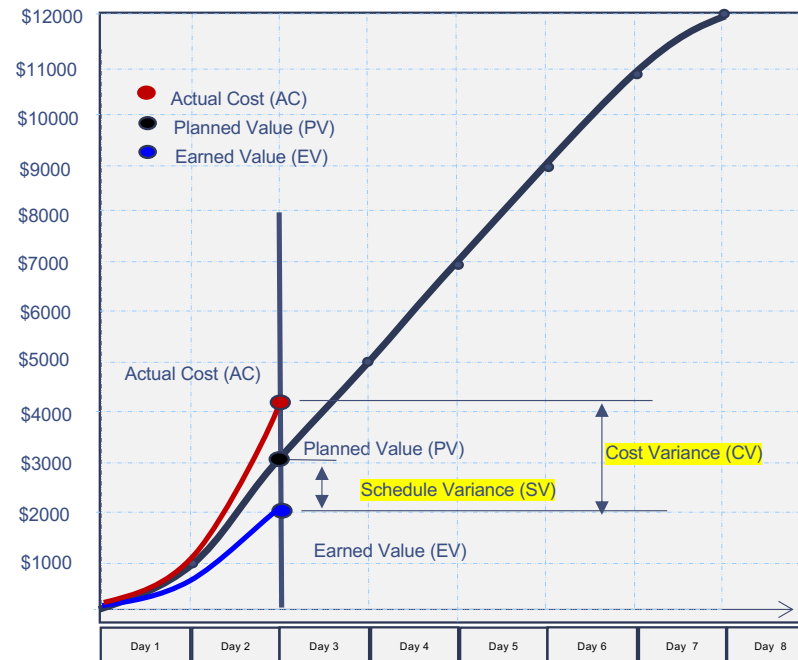
Day	Work	Person required (no.)	Planned Rate (\$)	Planned Cost (\$)	Actual Rate (\$)	Actual Cost (\$)
1	Installation	1	\$1000	\$1000	\$1100	\$1100
2	Installation	2	\$1000	\$2000	\$1500	\$3000
3	Installation	2	\$1000	\$2000		
4	Installation	2	\$1000	\$2000		
5	Installation	2	\$1000	\$2000		
6	Installation	2	\$1000	\$2000		
7	Final handover	1	\$1000	\$1000		

How do we represent this information?



Earned Value Management (EVM) Example

Activities	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Set Up	■						
Planned Cost	\$1000						
Install		■	■	■	■	■	
Planned Cost		\$2000	\$2000	\$2000	\$2000	\$2000	
Handover							■
Planned Cost							\$1000
Cum Cost (PV)	\$1000	\$3000	\$5000	\$7000	\$9000	\$11000	\$12000
Rate Budget	\$1000 person per day						
Work Done	100%	50%					
EV	\$1000	\$1000					
Cum (EV)	\$1000	\$2000					
Rate Actual	\$1100 1 st day and \$1500 person per day						
Actual Cost AC	\$1100	\$3000					
Cum (AC)	\$1100	\$4100					



$$\begin{aligned} \text{Cost Variance (CV)} &= \text{EV} - \text{AC} \Rightarrow \$2000 - \$4100 = -\$2100 \\ \text{Schedule Variance (SV)} &= \text{EV} - \text{PV} \Rightarrow \$2000 - \$3000 = -\$1000 \end{aligned}$$

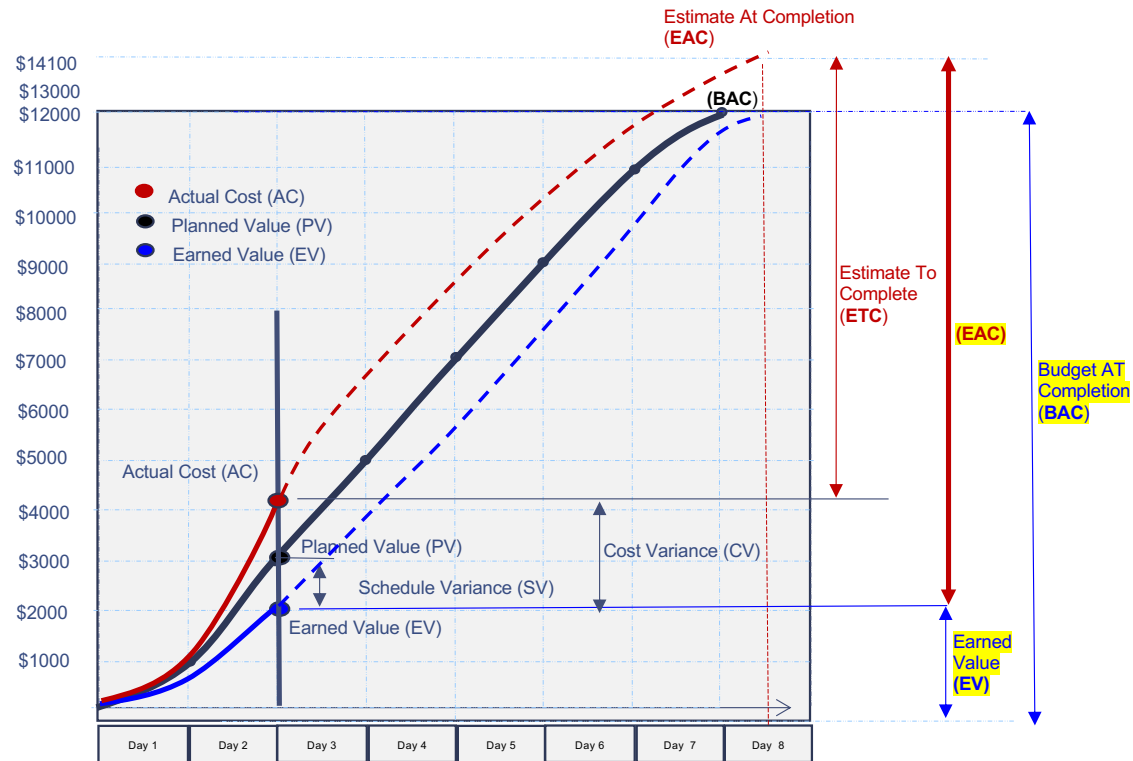
+ve value is good as it means you have earned more than planned.
-ve value is BAD as it means you are falling behind.

$$\begin{aligned} \text{Cost Performance Index (CPI)} &= \text{EV} / \text{AC} \Rightarrow \$2000 / \$4100 = 0.48 \\ \text{Schedule Performance Index (SPI)} &= \text{EV} / \text{PV} \Rightarrow \$2000 / \$3000 = 0.67 \end{aligned}$$

< 1 is not good, means you are behind



Forecasting



$$EAC = AC + (BAC - EV) \quad \text{or} \quad AC + ETC$$

$$= \$4,100 + (\$12,000 - \$2,000)$$

$$= \$14,100$$

Estimate AT Completion => **EAC**
 Budget AT Completion => **BAC**
 Estimate To Complete => **ETC**



Project Topic and Approach

AI in Project Management



AI in Project Management:

- Criteria to select a project for group assignment:
 - Can have either or all phases of project covered in the lecture
 - Initiation,
 - planning,
 - executing,
 - monitoring & control,
 - close-out.
- Must draw a workflow of the current situation and proposed improvement.
- **Improve process** for a project by implementing AI tool of your choice. (Example: collating of notes of meeting, preparation of EVM, sorting the team profiles, etc.).



AI in Project Management: Where can it help a Project Manager?

INITIATING

- Project Business Case development.
- Review and analyze the contract document.
- Client supplied technical and commercial documents compile and analysis.
- Project Charter development.

PLANNING

- Project Management Plan preparation and updating.
- Project specific procedures.
- Project compliance to contract, local regulations, etc.

EXECUTING

- Team selection (Review of applications).
- Project schedule development and updates.
- Earned Value Management (EVM) and Forecasting.
- Budget vs Actual performance analysis.
- Notes of meeting automation, summary, issue to the stakeholders.
- Issue regular communication with the stakeholders. (Emails, Project Updates, Newsletters, etc.).
- Periodic project reports (weekly / monthly / quarterly).
- Project Quality and Inspection (Plan and Actual performance / analysis).
- Project Risk Analysis (Plan and Actual performance / analysis).
- Project Change Management and its impact on project. (Plan and Actual performance / analysis).
- Project procurement management (vendor, sub-con selection, tender comparison, recommendation, performance analysis, etc.).
- Expenses reporting.

MONITORING & CONTROL

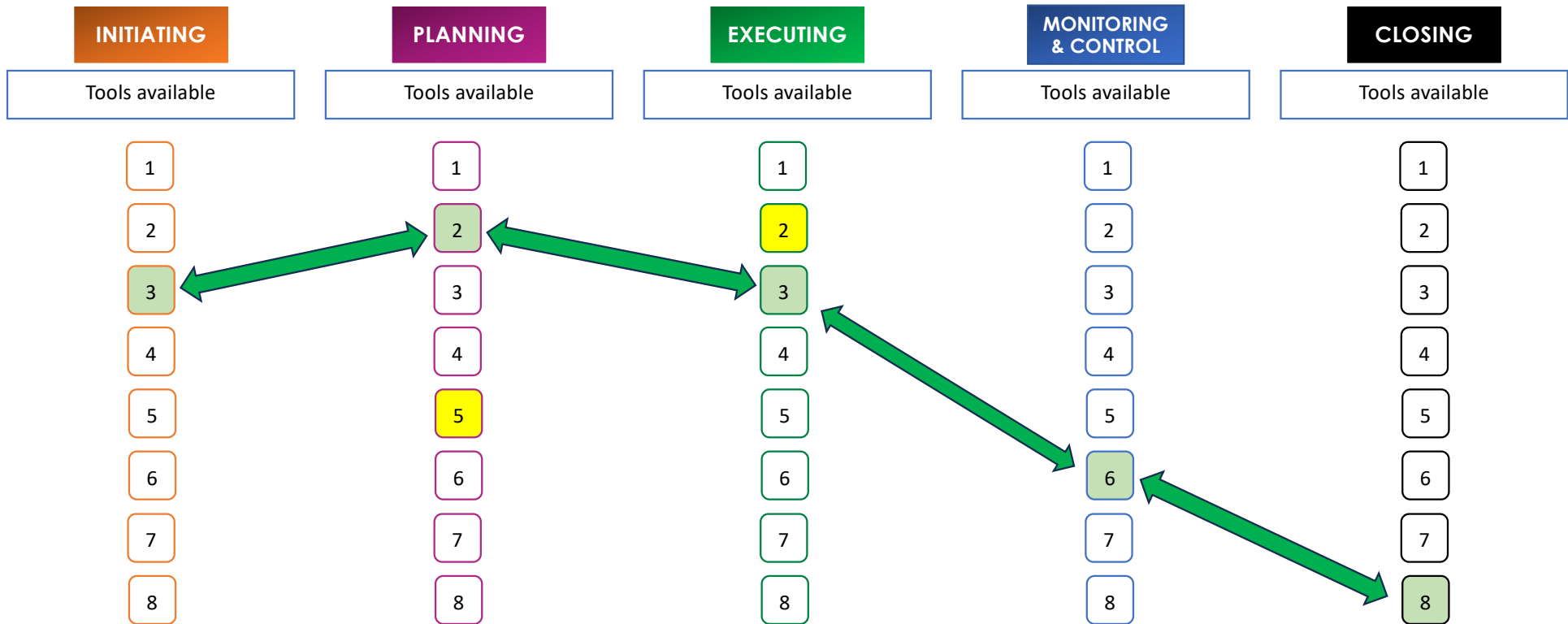
CLOSING

- Planned vs Actual Data analysis (Scope, Schedule, Cost, Resources).
- Changes and rework analysis.
- Quality/ Safety/ Compliance/ Testing/ Deployment/ Handing over, performance and analysis.
- Vendors / Sub-contractors/ Suppliers performance analysis.
- Team performance analysis.
- Lessons mapping and analysis.
- Project close-out documentation.

All of the above mentioned can be improved using AI Tools

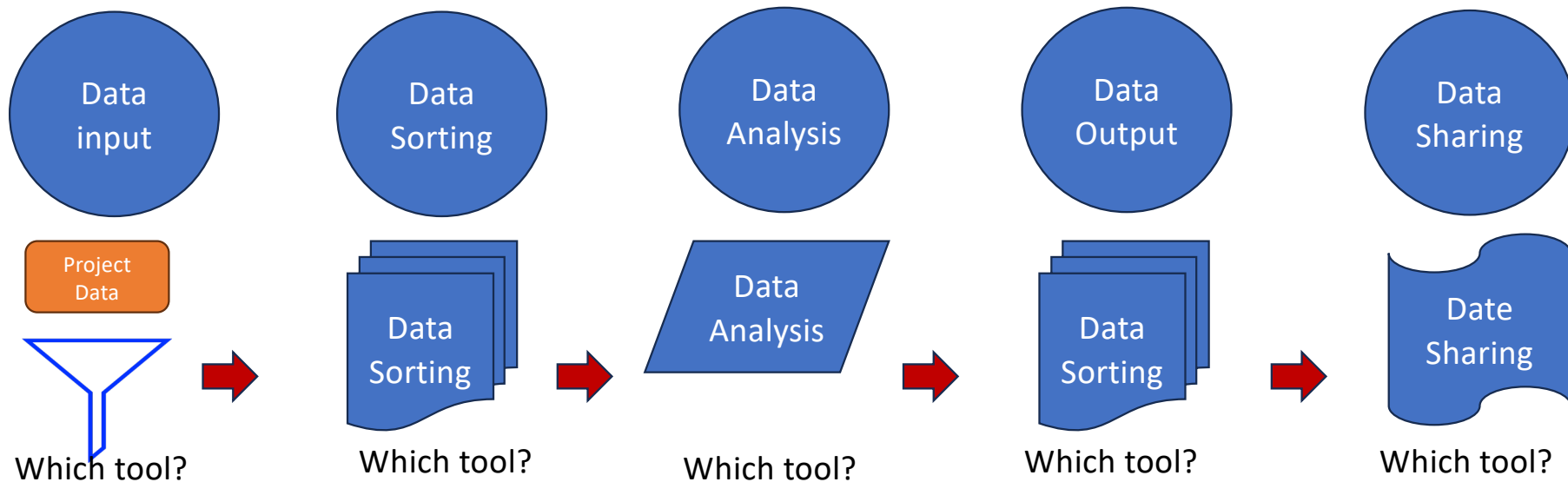


AI in Project Management: Select the right AI Tools





AI in Project Management: **Workflow**





AI in Project Management:

- Use the consultation session regarding the group project.
- AI Tools can be of your choice.
- Marking will have the rubrics to include:
 - Group report (content and quality).
 - Workflow snapshot.
 - Proposed AI tools for the solution to the workflow.



AI in Project Management:

- Use the project report template provided.
- Please do not delete the headings and bullets. (please mark N.A, if any of them is not applicable for your selected project).
- Use the guidelines provided for preparation.



Group Project: What Next?

AI in Project Management



Topic understanding and to set a path to work on.

Client wish to implement AI Tools
in Project Management

Client

- You have project topic selected.
- Your inputs:
 - Workflow with **project steps** explained.
 - **Identify the areas** where AI Tools can be implemented in Project Management Processes.
 - **Effort needed** to develop and implement the AI solutions.
 - **Proposed Time / Schedule** to execute the project.
 - Any other **value-added** pointers.

Your Group Represents as Consultant
and expected to advise the client.

Consultant



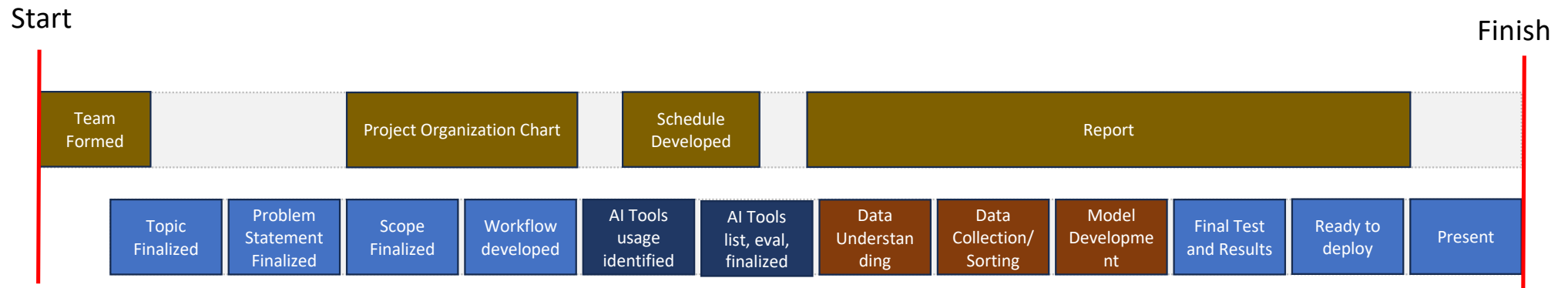


Your Group Project Progress?

AI in Project Management



Group Project Approach





Your Group Project Progress?

Actual Work
Progress

1. Workflow understanding.
2. Workflow developed and finalized.
3. AI Tools usage understood and agreed.
4. AI Tools list is finalized.
5. Proposed project schedule, cost, resource requirements are developed.
6. Other relevant details progress, risk identification, challenges, lessons, opportunities, etc.

7. Following are applicable if you are also developing a demo with some dummy data:
 - Data collection.
 - Data understanding.
 - Data sorting and preparation.
 - Model development.
 - Model testing.
 - Testing complete.
 - Results verification and validation.



Project Report Writing Progress

AI in Project Management



Group project report preparation progress

Report
Preparation
Progress

- Executive Summary – Better to write at last after the whole project report is completed.
- Project Introduction / Description / Problem Statement.
- Project Management and implementation:
 - Scope of work.
 - Project Organization chart.
 - Workflow diagram for your project.



Group project report preparation progress

Report
Preparation
Progress

- Project Schedule.
 - Gantt Chart for the project with key milestones.
- Project Cost:
 - Cost of AI Tools and other resources. (Cloud, server, etc.)
 - Cost of people for development.
 - Cost of Vendors and Suppliers (as may be applicable).
 - Cost of testing, deployment.
 - Cost of training staff.
 - Others.
- Overall performance of the project. Plan vs Actual.



Group project report preparation progress

Report
Preparation
Progress

- AI Tools used on the project.
 - List all the tools used and its purpose on this project.
 - Source of the AI tools (Vendors / Suppliers/ 3rd party installers as may be applicable).
 - License & Cost of each AI tool.
 - Schedule to implement these tools.
 - Training needs to apply these tools.



Group project report preparation progress

Report
Preparation
Progress

- AI Tools usage / implementation:
 - Understanding: why did you select them?
 - What are other 3 best choices available in the market (if any)?
 - Explain data understanding / data preparation approach and requirements for each AI Tool proposed.



Group project report preparation progress

Report
Preparation
Progress

- Model development / evaluation and deployment.
 - Explain model development for this whole project.
 - Draw a workflow as may be applicable.
 - Explain evaluation approach and results.
 - Deployment strategy and results.



Group project report preparation progress

Report
Preparation
Progress

- Benefits of implementing these AI Tools on your project.
 - Explain for the whole project.
- Challenges of implementing these AI Tools on your project.
 - Explain for the whole project.
- Risks of implementing these AI Tools on your project.
 - Explain for the whole project.



Group project report preparation progress

Report
Preparation
Progress

- Lessons learned while implementing these AI Tools on your project.
 - Explain for the whole project.
- Any other points you wish to share.



Group project report preparation progress

Report
Preparation
Progress

- Workflow snapshots/ screenshots/ coding/ inputs/ results.
 - Explain for the whole project.
 - Attachments.
 - Problem statement.
 - Input data
 - Output



Group project report preparation progress

Report
Preparation
Progress

- All slides reviewed and ready to present.
 - Check the slides.
 - All slides are marked with person presenting.
 - All slides' data is correct.

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Noise Cancellation
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6TH & 7TH



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55 Smartwatch

8TH to 52ND



SGD \$30
Grab Voucher



What to do before next consultation?



Get Ready for your draft work

- Any group can be asked to about their group work progress.
- It could be discussion on their project report draft.
- At random any of the groups can be picked to share.



What to do before next tutorial class?



Tutorial Classes

- Ensure you have **watched the relevant lecture(s)**.
- Familiarize yourself with
 - **Project Cost related concepts on a project**
 - **EVM (Earned Value Management)**
- Please **bring your laptop** to participate in class exercises/ activities.



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End of class



Nikon D850, 44mm, f/14, 1/400s